

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

Q.1. What is motional emf? Derive its formula $\mathcal{E} = -BDV$? Derive a formula to calculate mechanical power $P = B^2 D^2 V^2 / R$ to be exerted on a loop of wire of width D and resistance R when it is dragged with velocity V inside and at right angle to the magnetic field.

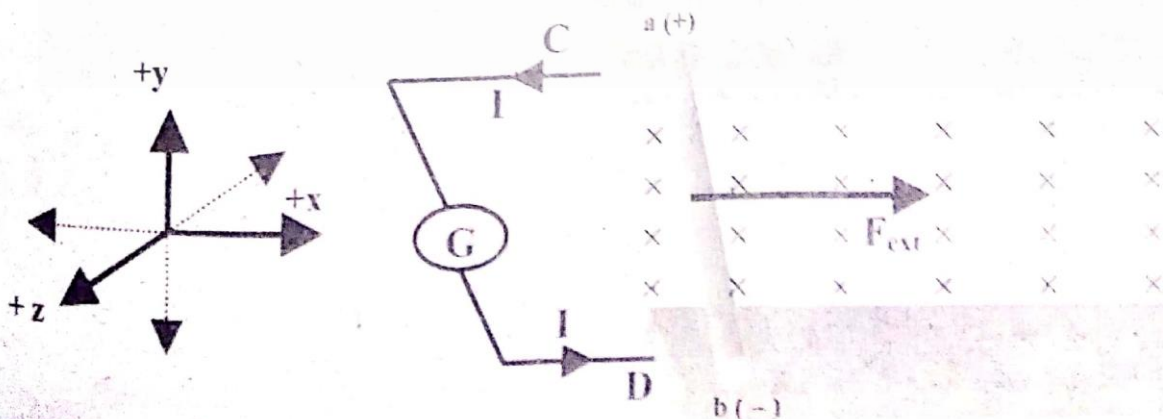
MOTIONAL EMF

The emf induced in a conductor when it is moved across a magnetic field is called motional emf.

FORMULA

Consider a conductor having length D is placed on a conducting rail whose ends C and D are connected with a galvanometer. A magnetic field $\vec{B} = -B \hat{k}$ is applied perpendicularly which is directed into plane of rail and denoted by $\times$.

The galvanometer shows deflection when rod is moved to and fro over the rail with constant velocity. It means emf is induced in a circuit. The charge $+q$ inside the rod moves with velocity $\vec{V} = V \hat{i}$ when rod is moved across the magnetic field towards right and experiences a magnetic force given as



$$\vec{F}_m = q (\vec{V} \times \vec{B})$$

$$\vec{F}_m = -q (V \hat{i} \times B \hat{k}) = qVB \hat{j}$$

The magnetic force acting on charge $+q$ is along $+y$ -axis. This magnetic force pushes $+q$ charges towards end a. The magnetic force acting on $-q$ charge in rod will be along $-y$ -axis according to right hand rule. Therefore, magnetic force pushes $-q$ charge towards end b. Hence electric field is induced in the rod along $+y$ -axis. The value of this electric field is given as

$$\vec{E} = \frac{\vec{F}_m}{q} = \frac{qVB \hat{j}}{q}$$

$$\vec{E} = VB \hat{j}$$

The magnitude of electric field is given as

$$E = VB$$

The electric field and emf are related as

$$\varepsilon = \int_D \vec{E} \cdot d\vec{r} = \int E dr \cos\theta$$

$$\varepsilon = \int_0^D VB dr$$

$$\varepsilon = VBD$$

INDUCED CURRENT

The amount of current induced in the loop having resistance R due to motional emf is given as

$$I = \frac{\varepsilon}{R} = \frac{VBD}{R}$$

The direction of induced current in the loop is such as it opposes the cause producing it. The direction of induced current in the loop is anticlockwise according to Lenz's law. Now rod having length D carrying current I is present in applied magnetic field $\vec{B} = -B \hat{k}$. The magnetic force $\vec{F}_L$ acting on the rod having length vector $\vec{L} = D \hat{j}$ is given as

$$\vec{F}_L = I (\vec{L} \times \vec{B})$$

$$\vec{F}_L = -IDB (\hat{j} \times \hat{k})$$

$$\vec{F}_L = IDB (-\hat{i})$$

The magnetic force acting on current carrying rod is in opposite direction to applied external force.

The magnitude of this magnetic force is given as

$$F_L = I D B$$

$$F_L = \frac{V B D (D B)}{R} = \frac{V B^2 D^2}{R}$$

MECHANICAL POWER

The rate of doing working is called mechanical power and given as

$$P_m = F_L V$$

$$P_m = \left(\frac{V B^2 D^2}{R} \right) V$$

$$P_m = \frac{V^2 B^2 D^2}{R} \text{ ----- (1)}$$

ELECTRICAL POWER

The rate of energy dissipated in the loop having resistance R due to induced current I is called electrical power and given as

$$P_e = I^2 R$$

$$P_e = \left(\frac{V B D}{R} \right)^2 R$$

$$P_e = \frac{V^2 B^2 D^2}{R} \text{ ----- (2)}$$

CONCLUSION

The above eq(1) and eq(2) gives $P_m = P_e$. It means mechanical power is completely converted into electrical power. The motional emf is the basic principle of AC generator which converts mechanical energy into electrical energy.

Q.2 State and explain Faraday's law of electromagnetic induction for a coil of N turns by giving the significance of negative sign. Write its integral and differential form?

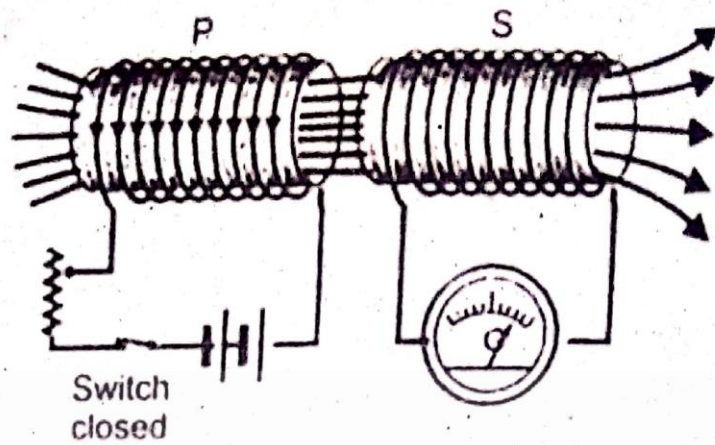
FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

The Faraday's law of electromagnetic induction states that whenever magnetic flux changes in a coil or loop, emf is induced in it which lasts only for the time, the flux change continue. The magnitude of induced emf is directly proportional to rate of change in magnetic flux.

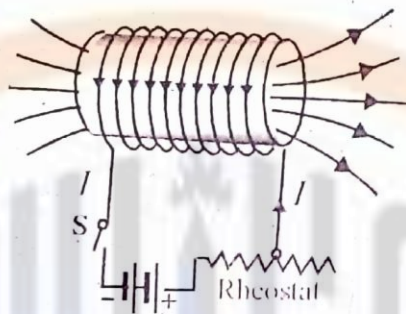
EXPLANATION

The emf is induced in a coil or loop due to change in magnetic flux. The magnetic flux in a coil or loop can be changed by number of ways.

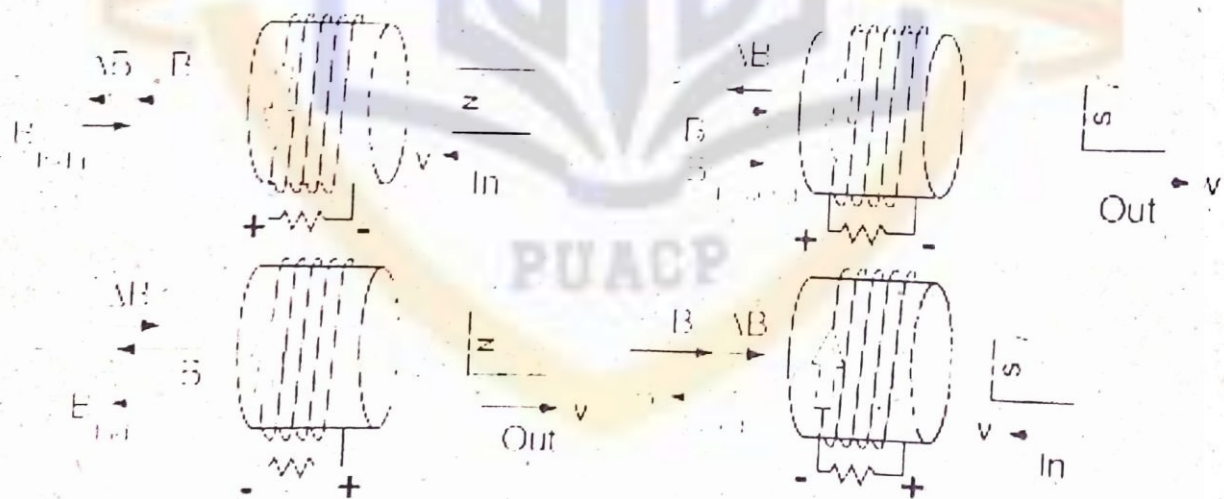
1 Mutual induction



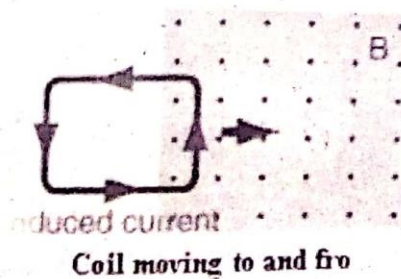
2 Self-induction



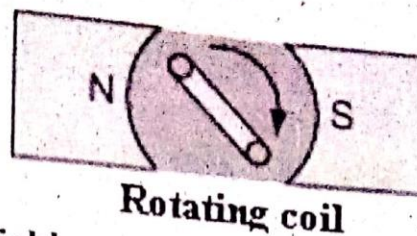
3 The to and fro motion of magnet in a coil or loop



4 The to and fro motion of coil in magnetic field

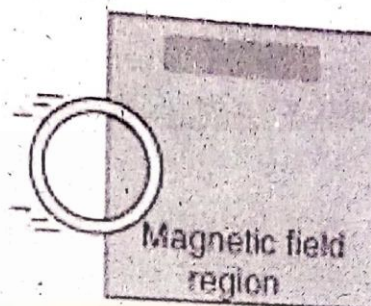


6 Rotation of a coil or loop in magnetic field



Rotating coil

6 Changing area in magnetic field



MATHEMATICAL FORM OF FARADAY'S LAW

Consider a conductor having length L is placed on a conducting rail as shown in fig. The ends of conducting rail are connected with galvanometer. The magnetic field B is directed into plane of paper and denoted by $(\times)$

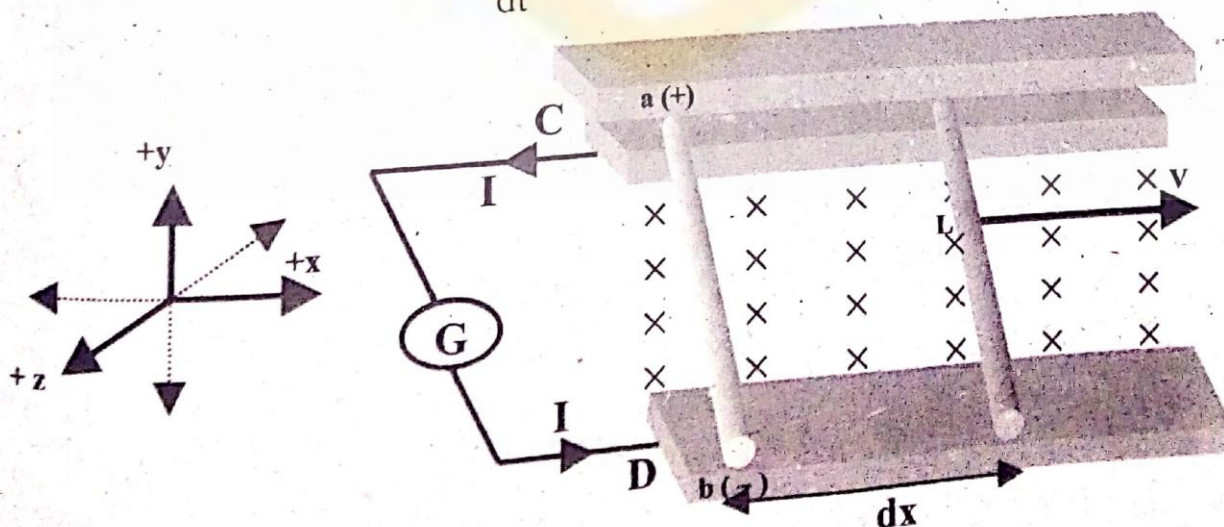
The galvanometer shows no deflection when rod is at rest. It means emf is not induced in rod. The galvanometer shows deflection when rod is moved to and fro across the magnetic field. It means emf is induced in loop due to change in magnetic flux. This deflection of galvanometer shows emf is only induced when flux is changing in loop. The emf vanishes as flux change stops.

The emf induced in rod when it is pulled towards right with constant velocity V is given as

$$\varepsilon = -VBL$$

The velocity V of rod which travels distance dx in time dt is

$$V = \frac{dx}{dt}$$



Put this value in above eq

$$\varepsilon = - \left(\frac{dx}{dt} \right) BL$$

$$\varepsilon = - \frac{B da}{dt} \quad [L dx = da]$$

$$\varepsilon = - \frac{d\phi_m}{dt}$$

The induced emf when loop has N turns

$$\varepsilon = -N \frac{d\phi_m}{dt} \quad \text{----- (1)}$$

The negative sign means direction of induced emf is such as it opposes the cause producing it. The magnetic flux $d\phi_m$ is given as

$$d\phi_m = \vec{B} \cdot d\vec{a}$$

The net magnetic flux is

$$\phi_m = \int \vec{B} \cdot d\vec{a}$$

Put this value in eq (1)

$$\varepsilon = -N \frac{d}{dt} \int \vec{B} \cdot d\vec{a} \quad \text{----- (2)}$$

INTEGRAL FORM OF FARADAY'S LAW

The mathematical form of Faraday's law is

$$\varepsilon = -N \frac{d}{dt} \int \vec{B} \cdot d\vec{a} \quad \text{----- (3)}$$

The emf and electric field are related as

$$\varepsilon = \int \vec{E} \cdot d\vec{r} \quad \text{----- (4)}$$

Comparing eq (3) and eq (4)

$$\int \vec{E} \cdot d\vec{r} = -N \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

For N = 1

$$\int \vec{E} \cdot d\vec{r} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

This is called integral form of Faraday's law

DIFFERENTIAL FORM OF FARADAY'S LAW

Since

$$\int \vec{E} \cdot d\vec{r} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a} \quad \text{----- (5)}$$

The Stoke's theorem is

Comparing eq. (5) and (6)

$$\int \vec{E} \cdot d\vec{r} = \int (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} \quad \dots \dots \dots (6)$$

$$\int (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

$$\int \left(\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} \right) \cdot d\vec{a} = 0$$

Since $d\vec{a}$ cannot be zero so

$$\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} = 0$$

$$\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}$$

This is called differential form of Faraday's law of electromagnetic induction.

Q.3 State and explain Lenz's law and show that it is in accordance with law of conservation of energy.

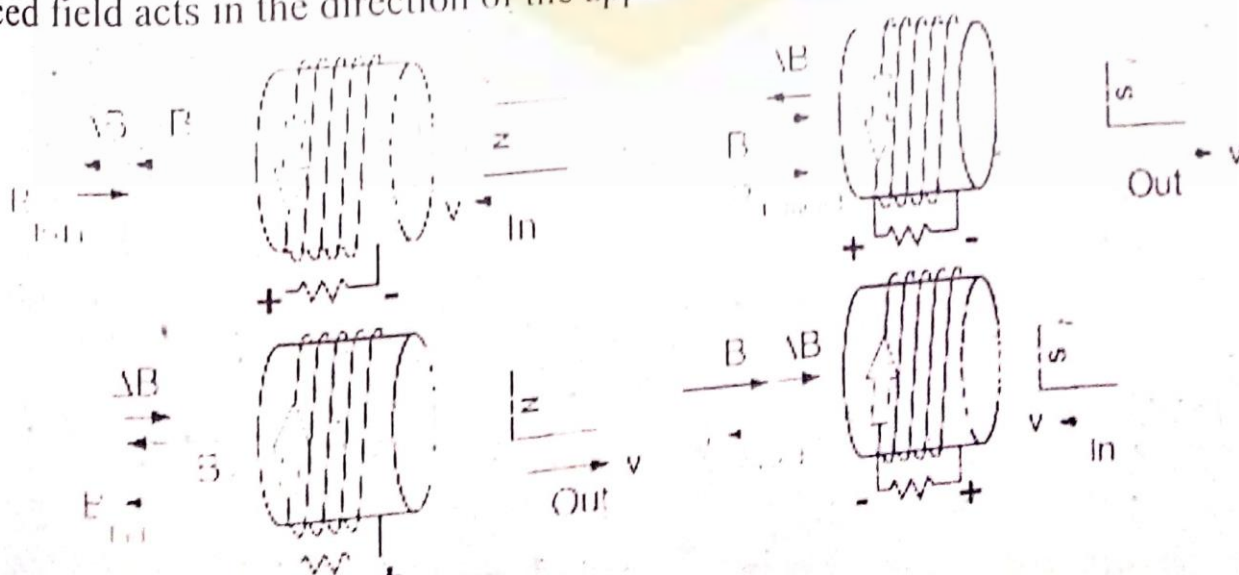
LENZ'S LAW

The Lenz's law states that the direction of induced current in a loop is such as it opposes the cause producing it.

EXPLANATION

The polarity of the induced emf which is generated by a change in magnetic flux according to Faraday's law is such that it produces a current whose magnetic field opposes the change which produces it.

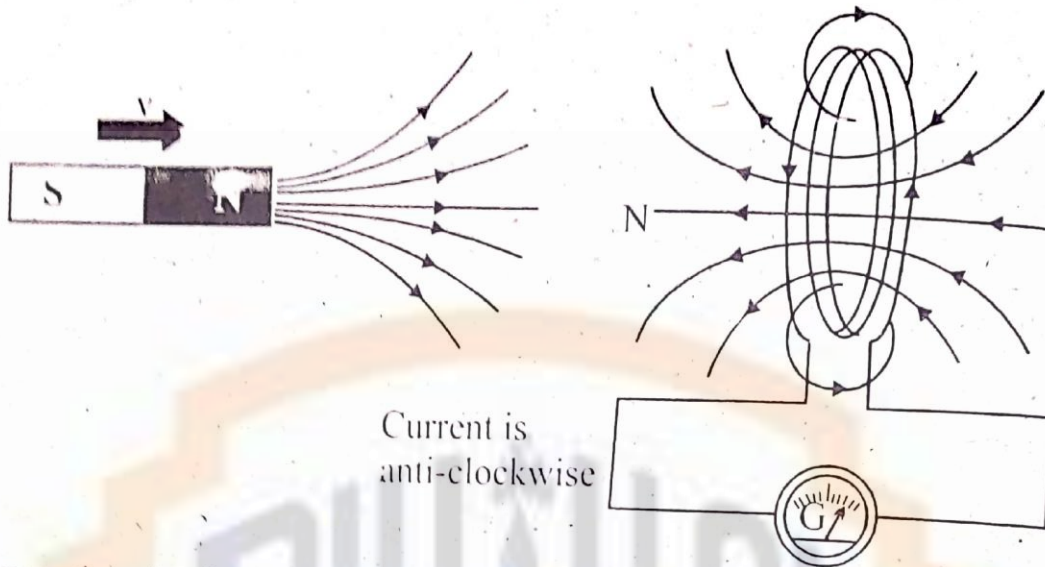
The induced magnetic field inside any loop of wire always acts to keep the magnetic flux in the loop constant. In the examples below, if the magnetic field B is increasing, the induced field acts in opposition to it. If magnetic field is decreasing, the induced field acts in the direction of the applied field to try to keep it constant.



EXPLANATION OF NEGATIVE SIGN IN FARADAY'S LAW

According to Faraday's law emf induced in a coil due to change in magnetic flux is written as

$$\varepsilon = -N \frac{d\phi_m}{dt}$$



The negative sign in the formula is very important. It is about the direction of induced emf and explained by Lenz as

Consider a coil connected with a galvanometer and a bar magnet as shown in fig. The galvanometer shows deflection when bar magnet is moved to and fro in the coil. The galvanometer shows no deflection when magnet is at rest inside the coil. It means current is induced in the coil due to movement of bar magnet inside the coil.

The current induced in loop must flow either in clockwise direction or anticlockwise direction. The Lenz's law decides the direction of current.

The direction of induced current must be anticlockwise when north pole of bar magnet is moved towards the coil. In this situation face of coil facing bar magnet becomes north pole which repels north pole of bar magnet so motion of bar magnet is opposed.

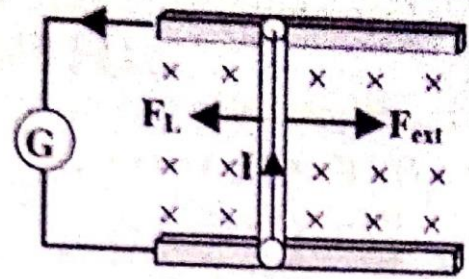
Similarly direction of induced current becomes clockwise when north pole of bar magnet is pulled out from coil. In this situation face of coil facing bar magnet becomes south pole which is attracted by north pole of bar magnet so motion of bar magnet is again opposed.

The "push" of the magnet is the "cause" that produces the induced current because magnetic field generated due to induced current opposes the motion of bar magnet. Similarly "pull" of the magnet is the "cause" that produces the induced current because magnetic field generated due to induced current opposes the motion of bar magnet.

LENZ'S LAW AND LAW OF CONSERVATION OF ENERGY

The Lenz's law is in accordance to law of conservation of energy because mechanical energy is converted into electrical energy when a conductor is moved across the magnetic field but total energy is constant.

Consider a conductor having length D is placed on a conducting rail whose ends are connected with a galvanometer. The magnetic field $\vec{B} = -B \hat{k}$ is directed into plane of paper and denoted by $\times$. The motional emf $\varepsilon = VBD$ is induced when rod is moved across the magnetic field towards right with constant velocity V .



The amount of current induced in the loop having resistance R due to motional emf is given as

$$I = \frac{\varepsilon}{R}$$

$$I = \frac{VBD}{R}$$

The direction of induced current in the loop is such as it opposes the cause producing it. The direction of induced current in the loop is anticlockwise according to Lenz's law. Now rod having length D carrying current I is present in applied magnetic field $\vec{B} = -B \hat{k}$. The magnetic force $\vec{F}_L$ acting on the rod having length vector $\vec{L} = D \hat{j}$ is given as

$$\vec{F}_L = I (\vec{L} \times \vec{B}) = -IDB (\hat{j} \times \hat{k}) = IDB (-\hat{i})$$

The magnetic force acting on current carrying rod is in opposite direction to applied external force. The magnitude of this magnetic force is given as

$$F_L = IDB = \frac{VBD(DB)}{R} = \frac{VB^2D^2}{R}$$

MECHANICAL POWER

The rate of doing working is called mechanical power and given as

$$P_m = F_L V = \left(\frac{VB^2D^2}{R} \right) V = \frac{V^2B^2D^2}{R} \quad \text{----- (1)}$$

ELECTRICAL POWER

The rate of energy dissipated in the loop having resistance R due to induced current I is called electrical power and given as

$$P_e = I^2 R = \left(\frac{VBD}{R} \right)^2 R = \frac{V^2B^2D^2}{R} \quad \text{----- (2)}$$

CONCLUSION

The above eq(1) and eq(2) gives $P_m = P_e$. It means mechanical power is completely converted into electrical power. Hence Lenz's law is in accordance with law of conservation of energy.

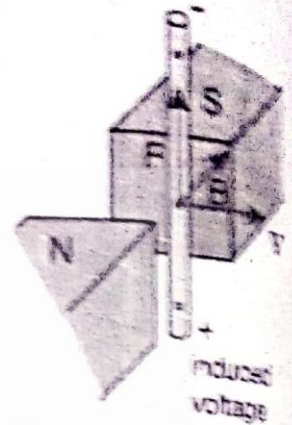
SHORT QUESTIONS

Q:1 In Faraday law of induction does the induced emf depend on the resistance of the circuit.

Ans: The induced emf in a circuit is given as

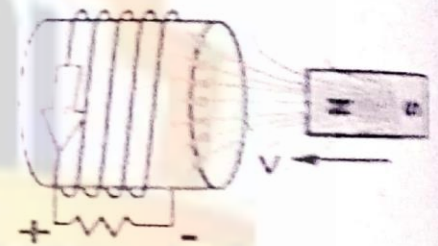
$$\varepsilon = -N \frac{d\phi_m}{dt}$$

Therefore, in Faraday law of induction, the induced emf does not depend on the resistance of the circuit.



Q:2 Is any emf induced in a long solenoid by a bar magnet that moves inside it along the solenoid axis.

Ans: Yes, emf is induced in a long solenoid by bar magnet that just enters into mouth of solenoid. The emf is not induced when bar magnet is completely inside the solenoid (having both north pole and south pole inside the solenoid). In this case net flux change is zero so



$$\varepsilon = -N \frac{d\phi_m}{dt} = 0$$

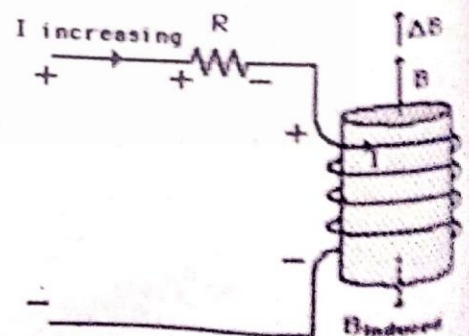
Q:3 Which quantities are measured in (a) weber (b) tesla-meter²? How are these quantities related with each other?

Ans: The magnetic flux is measured in units of weber. The magnetic flux is measured in units of tesla-meter². Both are related as weber = tesla-meter² because $\phi_m = B A$.

Q:4 Can back emf in an inductor be in the same sense as the emf of source.

Ans: The back emf in an inductor has the same sense as the emf of source in magnitude but direction of back emf and applied emf is opposite.

The increasing current in a coil of wire will generate a counter back emf which opposes the current.



Q:5 Show that Volt = Wb/s

Ans: $\frac{\text{Wb}}{\text{s}} = \frac{\text{T-m}^2}{\text{s}} \quad [\phi_m = B A]$

$$\frac{Wb}{s} = \left(\frac{N-s}{C-m} \right) \left(\frac{m^2}{s} \right) \quad [F_m = qVB]$$

$$\frac{Wb}{s} = \frac{N-m}{C}$$

$$\frac{Wb}{s} = \frac{J}{C}$$

$$[W = F_m d]$$

$$\frac{Wb}{s} = \text{Volt}$$

$$[\Delta V = W/q]$$

Hence relation is proved.

Q:6 Are induced emf and current different in any way from emf and current provided by battery connected to conducting loop.

Ans: The induced emf and current are same in magnitude to emf and current provided by battery connected to conducting loop but opposed in direction.

Q:7 Is the size of the voltage induced in a coil through which a magnet moves affected by the strength of the magnet, if so explain how.

Ans: The induced emf in a circuit is given as

$$\varepsilon = -N \frac{d\phi_m}{dt} = -N \frac{d(BA)}{dt}$$

The induced emf in a coil is directly proportional to magnetic field. The size of the voltage induced in a coil through which a magnet moves is affected by the strength of the magnet.

Q:8 Explain in your own words the difference between a magnetic field B and flux of magnetic field ϕ_m . Are they vector or scalar? What units may each be expressed? How are these units related?

Ans: The space or region around a magnet in which attraction or repulsion with other magnet is possible is called magnetic field. It is a vector quantity. Its SI unit is called tesla.

The total number of magnetic lines of force passing through a certain area is called magnetic flux. It is a scalar quantity. Its SI unit is called weber. The magnetic flux

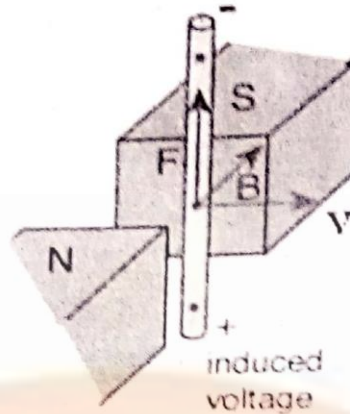
and magnetic field are related as $\phi_m = \vec{B} \cdot \vec{A}$

Q:9 You drop a bar magnet along the axis of a long copper tube. Describe the motion of the magnet and the energy interchanges involved. Neglect air resistance.

Ans: The emf is induced in copper tube when you drop a bar magnet along the axis of a long copper tube. The direction of induced emf is such as it opposes the motion of bar magnet. When magnetic force balances the gravitational force, the bar magnet attains the uniform velocity. The mechanical energy of bar magnet is converted into electrical energy.

Q:10 How you can summarize in one statement all the ways of determining the direction of an induced emf.

Ans: The direction of induced emf is that along which magnetic force acts on charge carrier in conductor.



Q:11 What do the symbols ϵ_0 and μ_0 stand for? How are they related with some constant in free space?

Ans. The relation between permittivity ϵ_0 of free space and permeability μ_0 of free space is given as

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} = 3.0 \times 10^8 \text{ m/s}$$

Where permittivity of free space is $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2$. The permeability of free space is $\mu_0 = 4 \times 10^{-7} \text{ Wb/A-m}$

Q:12 Give examples of how the flux linked by a coil can change due to stretching or compression of the coil.

Ans. The area changes when a coil is stretched or compressed. The magnetic flux is $\phi_m = B A$. Hence flux changes.

Q:13 Does the induced emf always act to decrease the magnetic flux through a circuit?

Ans. The emf is induced in the circuit due to change in magnetic flux and given by Faraday's law as

$$\epsilon = - N \frac{\Delta \phi_m}{\Delta t}$$

According to Lenz's law, the direction of emf always opposes the cause producing it. Therefore, emf acts to decrease the magnetic flux when magnetic flux is increasing in the circuit.

Similarly emf acts to increase the magnetic flux when magnetic flux is decreasing in the circuit. Hence the statement that induced emf always acts to decrease the magnetic flux through a circuit is not complete.

EXERCISE PROBLEMS

P:1 An automobile having a radio antenna 110 cm long travels 90 km/h in a region where magnetic field is 55 μT . Find value of induced emf.

Sol.

DATA

$$L = 110 \text{ cm}$$

$$V = 90 \text{ km/h} = 25 \text{ m/s}$$

$$B = 55 \mu\text{T}$$

$$\varepsilon = ?$$

CALCULATIONS

$$\varepsilon = V B L$$

$$\varepsilon = 25 \times 55 \times 10^{-6} \times 110 \times 10^{-2}$$

$$\therefore \varepsilon = 1.5 \text{ mV}$$

P:2 At a certain location the earth magnetic field has a magnitude of 42 μT and points downward at 57° to the vertical. Calculate the magnetic flux through a horizontal surface of area 2.5 m^2 .

Sol.

DATA

$$A = 2.5 \text{ m}^2$$

$$B = 42 \mu\text{T}$$

$$\theta = 57^\circ$$

$$\phi_B = ?$$

CALCULATIONS

$$\phi_B = B A \cos\theta$$

$$\phi_B = 42 \times 10^{-6} \times 2.5 \times \cos 57^\circ$$

$$\phi_B = 5.7 \times 10^{-5} \text{ Wb}$$

P:3 In a 100 MeV betatron, the orbit radius is 84 cm. The magnetic field in the region enclosed by the orbit rises periodically from zero to maximum average value $B = 0.80$ T in an interval of 4.2 ms. Calculate the average emf in weber per second.

Sol.

DATA

$$N = 1$$

$$KE = 100 \text{ MeV}$$

$$r = 84 \text{ cm}$$

$$dB = 0.80 \text{ T}$$

$$dt = 4.2 \text{ ms}$$

$$\varepsilon = ?$$

CALCULATIONS

$$\varepsilon = -N \frac{d\phi_B}{dt} = -N \frac{A dB}{dt}$$

$$\varepsilon = -N \frac{\pi r^2 dB}{dt}$$

$$\varepsilon = - \frac{\pi (84 \times 10^{-2})^2 (0.80)}{4.2 \times 10^{-3}} = -422 \text{ Wb/s}$$

P:4 The armature of a motor has 97 turns of area 190 cm^2 and rotates in a uniform magnetic field of 0.33 T . A Pd of 24 V is applied. If no load is attached and friction is neglected. Find the rotational speed at equilibrium.

Sol.

DATA

$$N = 97$$

$$A = 190 \text{ cm}^2$$

$$B = 0.33 \text{ T}$$

$$\varepsilon_0 = 24 \text{ Volt}$$

$$\omega = ?$$

CALCULATIONS

$$\varepsilon_0 = N A B \omega$$

$$24 = 97 \times 190 \times 10^{-4} \times 0.33 \times \omega$$

$$\omega = 40 \text{ rad/s}$$

P:5 A circular TV antenna has diameter of 11.2 cm. The magnetic field of TV signal is normal to the plane of loop. At one instant of time its magnitude is changing at the rate of 157 mT/s. Find the emf induced in the antenna.

Sol.

DATA

$$N = 1$$

$$d = 11.2 \text{ cm}$$

$$\frac{dB}{dt} = 157 \text{ mT/s}$$

$$\varepsilon = ?$$

CALCULATIONS

$$\varepsilon = -N \frac{d\phi_B}{dt} = -N \frac{A dB}{dt}$$

$$\varepsilon = -N \frac{\pi r^2 dB}{dt}$$

$$\varepsilon = -\pi (5.6 \times 10^{-2})^2 (157 \times 10^{-3})$$

$$\varepsilon = -1.5 \times 10^{-3} \text{ V}$$

P:6 A loop of antenna having area A and resistance R is perpendicular to a uniform magnetic field B. The magnetic field drops linearly to zero in a time interval Δt . Find an expression for the total energy dissipated in the loop.

Sol.

DATA

$$N = 1$$

$$\text{Area of loop} = A$$

$$\text{Resistance of loop} = R$$

$$\text{Magnetic field} = B$$

$$\text{Time interval} = \Delta t$$

$$dU_m = ?$$

CALCULATIONS

$$P = I \varepsilon = \frac{\varepsilon^2}{R}$$

$$P = \frac{1}{R} \left(N \frac{d\phi_m}{dt} \right)^2$$

$$\frac{dU_m}{dt} = \frac{1}{R} \left(A \frac{dB}{dt} \right)^2$$

$$\frac{dU_m}{dt} = \frac{A^2 B^2}{R dt} \quad [dB = B_2 - B_1 = B - 0 = B]$$

P:7 A square wire loop with 2.3 m side is perpendicular to a uniform magnetic field B , with half the area of the loop in the field. The loop contains a 2 V battery with negligible internal resistance. If the field varies with time according to $B = (0.42 - 0.87 t) \text{ T}$. Find the total emf in circuit.

Sol.

DATA

$$N = 1$$

$$a = 2.3 \text{ m}$$

$$A' = 2.3 \times 2.3 = 5.29 \text{ m}^2$$

$$V = 2 \text{ Volt}$$

$$B = (0.42 - 0.87 t) \text{ T}$$

$$\varepsilon = ?$$

CALCULATIONS

Half area of loop

$$A = 2.64 \text{ m}^2$$

$$\frac{dB}{dt} = -0.87 \text{ T/s}$$

The emf induced in loop

$$\varepsilon = -N \frac{d\phi_m}{dt}$$

$$\varepsilon = -N \frac{A dB}{dt}$$

$$\varepsilon = -(1)(2.64)(-0.87) = 2.30 \text{ V}$$

The emf induced in loop is opposite to emf of battery so total emf in circuit

$$\text{Net emf} = 2 + 2.3 = 4.3 \text{ V}$$

P:8 A circular loop made of a stretched conducting elastic material has 1.23 m radius. It is placed with its plane right angle to 785 mT magnetic field. When released, radius of loop starts to decrease at 7.5 cm/s. Calculate emf induced in loop.

Sol.

DATA

$$r = 1.23 \text{ m}$$

$$V = 7.5 \text{ cm/s}$$

$$B = 785 \text{ mT}$$

$$\varepsilon = ?$$

CALCULATIONS

$$\varepsilon = V B L$$

$$\varepsilon = V B (2\pi r)$$

$$\varepsilon = 7.5 \times 10^{-2} \times 785 \times 10^{-3} \times 2\pi \times 1.23$$

$$\varepsilon = 0.454 \text{ V}$$

P:9 A conducting wire of fixed length can be wound into N circular turns, used as armature of generator. To get the largest emf, what value of N would you choose?

Sol.

The length of wire = L

Maximum voltage = ϵ_0

$$\epsilon_0 = N A B \omega$$

The value of N must be reciprocal of area of loop

$$N = \frac{1}{A} = \frac{1}{\pi r^2}$$

But $L = 2\pi r$, This gives $r = \frac{L}{2\pi}$

Hence we can write as

$$N = 4 \frac{\pi}{L^2}$$

P:10 A conductor of length 0.5 m is sliding along a loop of 0.7Ω resistance with velocity 0.2 m/s. The loop is placed in magnetic field of 1.5 T. (a) Find the motional emf if the conductor moves perpendicular to field. (b) Current through the loop. (c) Electric power (d) Mechanical power.

Sol.

DATA

$$L = 0.5 \text{ m}$$

$$V = 0.2 \text{ m/s}$$

$$B = 1.5 \text{ T}$$

$$R = 0.7 \Omega$$

$$\epsilon = ?$$

CALCULATIONS

$$\begin{aligned} \epsilon &= V B L \\ \epsilon &= 0.2 \times 1.5 \times 0.5 = 0.15 \text{ V} \end{aligned}$$

$$I = \frac{\epsilon}{R} = \frac{0.15}{0.7} = 0.21 \text{ A}$$

$$\begin{aligned} P_m &= I_m V = ILB (V) \\ P_m &= 0.21 \times 0.5 \times 1.5 \times 0.2 = 0.032 \text{ W} \end{aligned}$$

$$P_e = I \epsilon = 0.21 \times 0.15 = 0.032 \text{ W}$$

(d)

P:11 A uniform magnetic field is normal to the plane of a circular loop 10.4 cm in diameter made of copper wire ($R = 1.12 \times 10^{-3} \Omega$). At what rate must the magnetic field change with time if an induced current of 9.66 A is to appear in the loop?

Sol.

DATA

$$d = 10.4 \text{ cm}$$

$$r = 5.2 \text{ cm} = 5.2 \times 10^{-2} \text{ m}$$

$$R = 1.12 \times 10^{-3} \Omega$$

$$I = 9.66 \text{ A}$$

$$\frac{dB}{dt} = ?$$

CALCULATIONS

$$\varepsilon = \frac{d\phi_m}{dt} = \frac{d(BA)}{dt} = A \frac{dB}{dt}$$

$$\frac{dB}{dt} = \frac{\varepsilon}{A} = \frac{IR}{\pi r^2}$$

$$\frac{dB}{dt} = \frac{9.66 \times 1.12 \times 10^{-3}}{\pi (5.2 \times 10^{-2})^2} = 1.27 \text{ T/s}$$

